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**Extensively-drug resistant (XDR) *Klebsiella pneumoniae*
associated with complicated urinary tract infection in
Northern India**

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Extensively-drug resistant (XDR) *Klebsiella pneumoniae* associated with complicated urinary tract infection in Northern India

Work was carried out at Enteric Laboratory, Department of Medical Microbiology,
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Running title XDR *Klebsiella pneumoniae* in UTI

Abstract

Klebsiella pneumoniae (Kp) associated with hospital acquired infections are extensively drug resistant (XDR), making treatment problematic. Understanding the genetic epidemiology of XDR -Kp can determine their potential to be hypervirulent (hv) through the presence of siderophores. We characterized genomes of 18 colistin-resistant XDR-Kp isolated from 14 patients with complicated urinary tract infection in an Indian healthcare facility.

18 organisms comprised STs: ST14 (9/18), ST147 (5/18), ST231 (2/18), ST2096 (1/18), and ST25 (1/18). Many patients in one ward were infected with the same ST, indicating a common infection source. Some patients had recurrent infections with multiple STs that were circulating in that ward, providing evidence for hospital transmission. Beta lactamase genes (*bla*_{CTX-M-1}, *bla*_{SHV}, and *bla*_{ampH}) were present in all isolates. *bla*_{NDM-1} was present in isolates 15/18, *bla*_{OXA-1} was present in isolates 16/18, *bla*_{TEM-1D} was present in 13/18 isolates, and *bla*_{OXA-48} was present in isolates 14, 19 and 30. Disruption of *mgrB* with various IS elements was responsible for colistin resistance in 6 isolates. The most common K type among these isolates was K2 (10/18). One XDR convergent hv-Kp ST2096 was associated with prolonged hospitalisation (*iuc+ybt+bla*_{OXA-1}+*bla*_{OXA-48}). Convergent XDR-hv-Kp detected has outbreak potential, warranting effective antimicrobial stewardship and infection control.

Introduction

There is growing concern regarding the disease burden associated with the ESKAPE group of pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*,

Acinetobacter baumannii, *Pseudomonas aeruginosa*, and *Enterobacter* species). These bacteria are frequently multi-drug resistant (MDR) and are a leading cause of hospital-acquired infections internationally (1). More specifically, *K. pneumoniae* is a major emerging pathogen and has been implicated in a wide variety of infections, including meningitis, pneumonia, sepsis, intra-abdominal infections, skin and soft tissue infections, and urinary tract infections (UTIs) (2). *K. pneumoniae* that are acquired in hospitals in South Asia are frequently extensively-drug resistant (XDR) or pan-drug resistant (PDR), making them highly problematic (3). An increase in XDR and PDR *K. pneumoniae* has become a global concern, and the World Health Organization (WHO) has placed carbapenem and third-generation cephalosporin-resistant *K. pneumoniae* on the priority list of pathogens requiring new antimicrobials (4).

The most dangerous form of *K. pneumoniae* are associated with invasive disease and were first isolated from liver abscesses; these invasive organisms are considered hypervirulent (hvKp) (5). HvKp variants are more commonly community-acquired, but their capacity to infect healthy individuals may also facilitate their transmission in healthcare facilities. Consequently, hvKp infections are becoming more prevalent and associated with increasing mortality (6). The molecular basis of hvKp is not implicit, but isolates that carry the salmochelin, enterobactin, and aerobactin siderophores appear to be better adapted for systemic spread in humans. Notably, the *iuc* locus (*iucABCD*), which encodes aerobactin, is located on a plasmid and can theoretically be horizontally transferred from one organism to another (6). Fortunately, hvKp isolates are generally susceptible to the majority of clinically relevant antimicrobials, but there are emerging reports of hvKp isolates with MDR phenotypes (7). The combination of extreme AMR phenotypes (MDR, XDR, and carbapenem resistance) and hypervirulence in *K. pneumoniae* represent a major threat to public health.

K. pneumoniae is the second most common cause of UTI in our tertiary care centre in Northern India, accounting for 25% of all UTIs; 13.9% of these are XDR (8). The antimicrobial susceptibility of these organisms determines that colistin is now the last resort drug for the treatment of XDR *K. pneumoniae* infections (9). However, the reported prevalence of colistin non-susceptibility in MDR *K. pneumoniae* of 18.5% in our centre is also high (8). The plasmid-mediated colistin resistance gene *mcr* and mutations in the two-component systems PmrA/B and PhoP/Q, and MgrB, a negative regulator of the PhoP/Q, lead to colistin resistance and compromise the potential efficacy of this vital drug (10-12)

With the potential health implications of complicated UTIs caused by MDR and XDR organisms, it is essential to better understand their epidemiology and genetic composition. Whole-genome sequencing (WGS) can help us identify both known and novel mechanisms of antimicrobial resistance, determine sequence types (STs), and aid epidemiological investigations through phylogenetics (12). Here, we characterized the genomes of 18 colistin-resistant XDR *K. pneumoniae* that were isolated from 14 patients with complicated UTI over a six-month period in a tertiary hospital in Northern India.

Materials and Methods

Study design

We conducted a retrospective study to characterize the genome sequences of 18 colistin non-susceptible uropathogenic XDR *K. pneumoniae* isolated from urine samples of 14 in-patients attending the Postgraduate Institute for Medical education and Research (PGIMER) in Chandigarh. The clinical details of the patients were recorded from the case files after obtaining an informed consent from the adult patients and parent(s)/legal guardians of all

cases under 18 years of age. This study was approved by the internal ethics review board of PGIMER.

Microbiological methods

Urine samples from patients suffering from UTI (Fever >100° F, burning micturition, and increased frequency of urination) admitted to PGIMER between July 2017 and January 2018 were processed according to the standard operating procedures of the Enteric Laboratory (8). Samples were inoculated on cysteine lactose electrolyte deficient (CLED) media; yellow colonies (lactose fermenting) indicative of *K. pneumoniae* were subjected to Matrix-assisted laser desorption/ ionization-time of flight mass spectrometry (MALDI-TOF) (Bruker Dalton, Germany). Those confirmed as *K. pneumoniae* were subjected to antimicrobial susceptibility testing (AST) and their clinical details were accessed from case records (8).

AST was performed using the Kirby-Bauer disc diffusion method for aminoglycosides [amikacin (30 µg), gentamicin (10 µg)], beta-lactams [piperacillin-tazobactam (100/10 µg)], carbapenems [ertapenem (10 µg), meropenem (10 µg), imipenem (10 µg)], cephalosporins [cefotaxime (30 µg), cefepime (30 µg), cefoparazone-sulbactam (75/30 µg)], co-trimoxazole (1.25/23.75 µg), quinolones [ciprofloxacin (5 µg), norfloxacin (10 µg), nalidixic acid (30 µg)], tetracyclines [tetracycline (30 µg), minocycline (30 µg)], nitrofurantoin (300 µg), tigecycline (15 µg) and fosfomycin (200 µg) (Becton, Dickinson and Company., Sparks, MD, USA) following the recommendations of the Clinical and Laboratory Standards Institute (CLSI) (13). MDR was defined as non-susceptibility to at least one agent in three or more antimicrobial categories and XDR was defined as non-susceptibility to at least one agent in all but two or fewer antimicrobial categories (i.e., bacterial isolates remaining susceptible to only one or two categories). XDR isolates were further tested using the CLSI broth microdilution

method to determine phenotypic resistance to colistin (Sigma Aldrich, USA) (14) Isolates with minimum inhibitory concentration (MIC) ≥ 2 $\mu\text{g/mL}$ against colistin were deemed as colistin non-susceptible. Colistin-susceptible *E. coli* ATCC 25922 and *P. aeruginosa* ATCC 27853 and the *mcr1*-positive *E. coli* NCTC 13846 (CCUG 70662, DSM 105182) with a colistin MIC of 4 mg/L were used as quality control (QC).

Whole-genome sequencing

Genomic DNA was extracted from all isolates using the Wizard genomic DNA purification kit (Promega, Madison WI, USA). Briefly, genomic DNA was purified from each isolate and quantified using a Qubit dsDNA high sensitivity assay (ThermoFisher, Waltham, USA). Libraries were prepared using the Nextera XT kit and sequenced using an MiSeq platform (2 x 250 bp, V2 500 cycles) (Illumina Inc., San Diego, CA, USA). The raw sequences were subjected to a quality check and analysis using an in-house pipeline, BacPipe (15). Briefly, reads were quality assessed, trimmed, and *de novo* assembled (SPAdes v 3.10.0) (16). The assemble sequences were subjected to MLST typing, annotation (Prokka v 1.12) (17), and antimicrobial resistance gene prediction using ARG-annot v3 (18). In order to determine the incompatibility type of the plasmids present in the sequences, the data were processed using ISFinder (19) and insertion element (IS) families were detected (20). Virulence genes were identified using VFDB (21) and capsular typing was performed using Kleborate v0.3.0 (<http://github.com/katholt/Kleborate>).

Phylogenetic Analysis

For phylogenetic comparison, all eighteen sequenced isolates along with a reference sequence (KPN528 with accession no CP020853.1) were passed through the RedDog pipeline V1beta.10.3 (<https://github.com/katholt/RedDog>). Phylogenetic relationships were inferred

using the Maximum Likelihood method based on a General Time Reversible model. A phylogenetic tree was constructed to scale, with branch lengths measured in the number of substitutions per site. SNP-sites v 2.5.1(22) and snp-dists v 0.7.0(23) were used to determine the genetic distance (Single Nucleotide Polymorphisms, SNPs) between paired sequenced isolates in addition to the reference genome (GenBank assembly accession number GCA_002156785.1) from the core gene alignment. The average nucleotide identity (ANI) in relation to the reference genome was assessed using fastani v 1.3(24).

Results

Baseline characteristics

Over the six-month sampling period 775 (3.1%) *K. pneumoniae* were isolated from 25,046 urine samples from patients with a clinically suspected UTI. Following antimicrobial susceptibility testing, 108 (13.9%) MDR *K. pneumoniae* isolates were identified. Eighteen of these *K. pneumoniae*, isolated from 14 patients, were non-susceptible to colistin, and were XDR. The majority of these *K. pneumoniae* were isolated from patients admitted to the urology wards (n=10), all of which had structural abnormalities of the urinary tract or had recently undergone urinary tract surgery (Table 1).

Antimicrobial resistance gene content and phylogenetic analysis

These 18 colistin resistant *K. pneumoniae* isolates were subjected to WGS. We observed the presence of a wide variety of AMR genes (41 individual genes) within these isolates. ST14 KPN528 was selected as a reference strain since ST14 was our most common ST. The beta-lactamase genes *bla*_{CTX-M-1}, *bla*_{SHV}, and *bla*_{Amp^H} were present in all isolates. (Figure 1) Since *bla*_{OKP-LEN} are common in *K. variicola*, *K. quasipneumoniae* and *bla*_{SHV} in *K. pneumoniae*, we calculated the genome average nucleotide identities (ANI) to with *Klebsiella pneumoniae*

reference genome (GenBank assembly accession number GCA_002156785.1) to accurately identify the species. The ANI is more than 98% in all isolates (range 98.72 to 99.97) confirming that our identity is accurate (24).

Pairwise comparison between sequences has been shown Figure 2. This is determined by the SNP distance matrix from core gene SNP alignment file. Core gene alignment file was obtained from the annotated assemblies, in addition to the reference genome (GenBank assembly accession number GCA_002156785.1) using Roaryv.3.13.0. Some of the strains have less SNP difference (7ST147 and 9ST147, one SNP difference); (2 ST 14, 26ST14, 2 SNP difference) ; (20ST14 and 24 ST14, 3 SNPs) ; (30ST231 and 19ST231, 3 SNPS) .

Furthermore, 15/18 sequences contained *bla*_{NDM-1}, of which 13/15 sequences also contained *bla*_{OXA-1}, and 2/15 contained *bla*_{OXA-48}. None of the isolates were KPC producers (Figure. 1). With respect to colistin resistance, the WGS data showed absence of *mcr1* to *mcr10* genes and mutations in the PhoPQ and PmrAB two-component systems. However, 7/18 isolates contained deletions/mutations in the *mgrB* genes, which included a disruption with *IS110* (most common)/*ISKpn26* and Cys28Gly (Table 2). This mutation has a deleterious effect and contributes to colistin resistance (25).

We identified five different STs; the most common was ST14 (n=9), followed by ST147 (n=5), ST231 (n=2), ST2096 (n=1), and ST25 (n=1) (Figure 1). We observed two phylogenetically distinct variants of ST147, of which three isolates exhibited an identical compendium of AMR genes and an indistinguishable plasmid profile.(Figure 1) Furthermore, two ST231 isolates additionally shared an identical plasmid with the same AMR genes (Table 2). Multiple STs were isolated from two patients (P3 and P12). Patient P3, with a bulbar urethral stricture, had recurrent UTIs with ST14 and ST147 and patient P12, with autosomal dominant polycystic kidney disease (ADPKD) and a right hydro-ureter, had multiple UTI

episodes caused by ST14, ST147, ST231 and ST25 (Table 1, Figure 3). We surmised that ST14 and ST147 were likely circulating concurrently in the urology ward.

As ST14 was the most common genotype, we analysed our ST14 sequences in depth. We detected a wide array (n=35) of AMR genes in these ST14 isolates, which were associated with resistance to several antimicrobial classes including aminoglycosides, trimethoprim, fluoroquinolones, phenicols, sulphonamides, streptomycin, tetracycline, fosfomycin, rifampin, 16S *rRNA* methylases, and beta lactams. Genes encoding resistance to aminoglycosides and fosfomycin (*fosA*) were present in all isolates (Figure 4). Additionally, the *bla*_{NDM-1}, *bla*_{OXA-1}, *bla*_{SHV} and *bla*_{CTX-M-1} were present in all isolates. We also detected *bla*_{TEM-1D} in four isolates (Figure 1).

Virulence gene analysis and capsular typing

We assessed the composition of virulence-associated genes. The *ybt* (yersinibactin) locus was the most common being present in 56% (10/18) of isolates and found in multiple STs [ST231 (n=2), ST2096 (n=1), ST14 (n=4), and ST147 (n=2) ST25(n=1)]. A single hypervirulent *K. pneumoniae* belonged to ST2096, a single locus variant of ST14. (Table 2) It was defined as hypervirulent based on presence of *iuc* (aerobactin), yersinibactin(*ybt*) and the hypermucoviscosity on culture (String test positive). It also contained enterobactins *fepG* and *entS* (26). The plasmid in this isolate also carried β -lactamases and carbapenemases (*bla*_{OXA-1} and *bla*_{OXA-48}). Consequently, we determined this ST2096 isolate to be a convergent XDR-hvKp variant. Many of the isolates (6/10) possessing *ybt* were also found to be colistin resistant, with corresponding mutations in *mgrB* in 5/10 isolates and deletion in 1 isolate (Table 2). All isolates with *ybt* locus also possess β lactamases *bla*_{SHV}, *bla*_{CTX-M-1}, 8/10 have *bla*_{TEM-1D}. 3/10 were carbapenemase (*bla*_{OXA-48}), and 9/10 were metallo- β lactamase (*bla*_{NDM-1}) producers (Table 2, Figure 1). Additionally, all isolates contained the *fepG* (iron enterobactin) and *entS* (enterobactin) siderophores, which also contribute to virulence (6). No other

hypermucous variants or isolates containing salmochelin or colibactin were identified. The most common K type among these isolates was K2 (10/18), followed by K10, K51, and K64 (Figure 1, Table 2).

Discussion

K. pneumoniae is the second most common organism associated with complicated with UTIs after *E. coli*, accounting for ~20% of cases at our centre (unpublished data). Crucially, the prevalence of MDR in *K. pneumoniae* is six times greater than observed in *E. coli* and UTIs caused by MDR *K. pneumoniae* are associated with increased morbidity (8). Colistin resistance is also emerging in India (27, 28), and we also identified organisms that were non-susceptible to colistin. Colistin resistance in *K. pneumoniae* is commonly associated with modification of lipid A, mutations in efflux pumps and the plasmid-mediated *mcr* genes (29). We did not detect any *mcr* genes or evidence for mutation in PmrA/PmrB, PhoP/PhoQ *ramA* or *ccrB*. We detected mutations in *mgrB*, which encodes a small transmembrane protein that mediates negative feedback on the PhoQ/PhoP system (30). Several studies have shown that mutations in *mgrB* (seen in 40% of cases observed here) lead to colistin resistance.(11) These mutations are thought to occur due to selective pressure of colistin usage, colonization, increased transmissibility and prolonged survival of these isolates. Other mechanisms like accumulation of capsular polysaccharide or efflux pumps may also contribute to colistin resistance in *K. pneumoniae*. (31)

ST14 was the predominant ST in this study. This is in accordance with the study by Long *et al.*, who also found ST14 to be common in India (32). A study from Vellore detected ST231, ST14 and ST147, which is again comparable to the distribution of STs observed here. By combining data from comparative analysis of ST14 isolates from aforementioned study in Vellore (31), the USA (32), and Pakistan (33), and our data, we observed that Indian isolates

possess more AMR genes than the organisms from other locations (Figure 4). Additionally, the ST14 isolates from India clustered into a single clade, indicating a common origin, which is concerning given that they have the capacity to cause UTIs and bloodstream infections. The majority of isolates here possessed enterobactin (*entS*), and half had siderophores such as yersinibactin (*ybt*), both of which contribute to the virulence (34). The iron scavenging ability of enterobactin can be limited by the binding of host lipocalin-2 to a ferric enterobactin. The presence of *ybt* aids in overcoming this defence mechanism and facilitates bacterial invasion (35, 36). Here, all isolates possessing *ybt* were either ESBL or carbapenemases producers. The *ybt* locus is mobilised by a self-transmissible integrative conjugative element (ICE), which can also carry additional virulence determinants (*iro*, *clb*) (35). We identified a single ST2096 *K. pneumoniae* with aerobactin (*iucABCD*), *ybt* and *bla_{OXA-1}* and *bla_{OXA-48}* that, that we defined as a convergent XDR-hypervirulent variant (Table 2). This organism was isolated from a patient with recurrent UTI and renal failure associated with urinary tract abnormalities. The patient was treated with aminoglycosides, cephalosporins, cotrimoxazole, and colistin during his 23 days of hospitalization. Despite antimicrobial therapy, recovery was prolonged and bacterial cultures continued to remain positive 10 days after corrective surgery. The presence of convergent hvKp have been reported from various parts of India (28) and Wyres *et al.* described Indian hvKp belonging to ST2096 and ST231 (7). Generally, hvKp is mostly confined to community transmission, but we speculate that the virulence plasmid can integrate into classical hospital XDR *K. pneumoniae* and trigger major outbreaks.

The main concern with highly resistant and hvKp inducing UTIs is that they are associated with high morbidity. Many patients exhibit clinical evidence of urosepsis, even in infections caused by non-hvKp. Here, we observed that several patients admitted to a single ward were infected with the same ST, indicating that these infections may have been acquired from a

common source. Concurrently, some patients had recurrent infections with multiple STs that were circulating on a specific ward at that time, providing further evidence for hospital transmission. As shown in Figure 2 very low genetic distance was observed between some isolates e.g. 7ST147 and 9ST147-one SNP; 2ST14 and 26ST14-two SNP; 20ST14 and 24ST14- three SNPs; 30ST231 and 19ST231- three SNPS. These appear to closely related, and may show nosocomial transmission.

Patients with recurrent UTIs often have prolonged hospital stays and have generally been exposed to multiple antimicrobials, including carbapenems and colistin (8). MDR *K. pneumoniae* that circulate in hospitals are a major concern as resistance genes are principally plasmid-mediated, and their ability to acquire further resistance genes gives them potential to become XDR and PDR. This scenario could lead to the development of more serious outbreaks associated with higher morbidity and mortality.

With the majority of our colistin-resistant isolates being ESBL and NDM-1 producers, there are limited options to treat these XDR infections (7). Therefore, we need new approaches to prevent and infections caused by these organisms. Formulating appropriate antimicrobial and infection control policies, compliance with hand hygiene, barrier nursing, and antimicrobial stewardship may aid in slowing the emergence of these highly resistant pathogens. Periodic surveillance of hospital wards to assess the circulating organisms and their antimicrobial susceptibility profiles should better support the design of empirical therapies.

Conclusions

This study provides new insight into the genes facilitating of AMR and virulence in *K. pneumoniae* associated with UTIs in India. Our data suggest that convergent XDR-hvKp is

present in our healthcare facility; we hypothesise that such organisms have outbreak potential, warranting more effective antimicrobial stewardship and better infection control strategies in India.

Declarations

Data Availability

Raw Illumina reads of *Klebsiella pneumoniae* isolates in this study have been deposited in the Sequence read archive (SRA), NCBI under the bio project PRJNA530687 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA530687>). Additionally, 13 published genomes of *K. pneumoniae* from study accession number PRJNA376412 (<https://doi.org/10.1128/mBio.00489-17>) (32) and PRJNA287968 (<https://doi.org/10.1371/journal.pone.0178888>) (37) were also included in the analysis.

Ethical approval

This work was a part of the MD Thesis of Dr. Parinitha Kaza, the study was conducted at PGIMER Chandigarh, after the approval of the Institute Ethics Committee of PGIMER reference no NK/3761/MD/386; vide letter no INT/IEC/2017/1009 dated 12.10.2017. The study was also approved from Institute Collaborative Research Committee of PGIMER vide letter no 79/227-Edu-18/4997 dated 12.12.2018. All methods were performed in accordance to the relevant guidelines and regulations of the Institute Ethics Committee.

Conflicts of interest

None

Consent to participate

Samples were included in the study after obtaining an informed consent from patients and patients and parent(s)/legal guardians of all cases under 18 years of age. Consent to publish was taken from patients at the time of sample collection as part of informed consent form.

Author contributions

PK Sample collection, processing, data analyses, manuscript writing

XBB Whole genome sequencing, MLST

JM Phylogenetic and MLST analyses, manuscript writing

NS SNP and ANI analysis

SB Manuscript writing and editing

TN Capsular typing of *K. pneumoniae*, virulence determinants analyses

RSM clinical details

BM Editing manuscript

NT Conceptualization, Supervision, manuscript writing and editing

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Figure legends

Figure 1. Phylogenetic structure of colistin resistant *K. pneumoniae* isolates from this study.

A phylogenetic tree of *K. pneumoniae* isolates from this study was constructed based on core SNP alignment using RedDog pipeline and viewed on iTOL with a heat-map showing AMR genes (red) and plasmids replicons (blue).

Figure 2. Pairwise comparison between sequences of *K. pneumoniae*

Figure 3. Patients with recurrent UTI due to colistin resistant *K. pneumoniae*

A line diagram of multiple *K. pneumoniae* STs isolated from two patients showing their colistin MIC on x-axis and no of days on y-axis. Day 0 is day of 1st culture positivity

Figure 4. Resistance profile of ST14 isolates described in different studies compared with our isolates

Key AG= aminoglycosides, FOS= fosfomycin, MACRO= macrolides, QNL= quinolone, RIF= rifampicin, SUL= sulphonamide, TET= tetracycline, TMP= trimethoprim

Table 1: Patient clinical details

Clinical Details									
Patient code	Age (Y)	Sex	Ward	Diagnosis	Abnormality/ surgery of urinary tract	Days of hospital stay	No. of Episodes of UTI	ST	Isolate No.
P1	46	F	Urology ward	Invasive devices	yes	22	2	ST14	1
P2	40	M	Urology ward	Metabolic disorders	yes	15	1	ST231	30
P3	29	M	Urology ward	Invasive devices	yes	16	1	ST14, ST147	2, 7
P4	6	M	Urology ward	Metabolic disorders	yes	23	1	ST2096	14
P5	4	M	Paediatric Surgery	DJ stent	yes	16	0	ST14	13
P6	48	F	Urology ward	Invasive devices	yes	22	1	ST147	16
P7	33	M	Urology ward	Metabolic disorders	yes	25	3	ST14	20
P8	61	F	Urology ward	Metabolic disorders	yes	15	0	ST147	17
P9	11	M	Paediatric Surgery	DJ stent	yes	90	0	ST14	21
P10	5	M	Paediatric Surgery	DJ stent	yes	8	5	ST14	24
P11	7	M	Paediatric Surgery	DJ stent	yes	20	5	ST14	25
P12	64	M	Urology ward	Renal Parenchymal disease	yes	20	4	ST25, ST14, ST147, ST231	15, 26, 9, 19
P13	30	M	Urology ward	Metabolic disorders	yes	58	2	ST14	28
P14	34	M	Urology ward	Metabolic disorders	yes	25	8	ST147	27

Table 2. The distribution of AMR and virulence determinants in sequenced organisms

Organism ID	ST	Yersinia-bactin	YbST	K type	Aero-bactin/Enterobactin	Mutation in <i>mgrB</i> gene	<i>bla</i> _{CTX-M-1}	<i>bla</i> _{NDM-1}	<i>bla</i> _{OXA-48}
1	ST14	<i>ybt 14; ICEKp5</i>	140	KL2	<i>fepG, entS</i>	disrupted by IS110 element at 129th bp	+	+	-
2	ST14	<i>ybt14; ICEKp5</i>	140	KL2	<i>fepG, entS</i>	nil	+	+	-
13	ST14	-	0	KL2	<i>fepG, entS</i>	disruption by ISKpn26 element at 74th bp	+	+	-
20	ST14	-	0	KL2	<i>fepG, entS</i>	nil	+	+	-
21	ST14	<i>ybt 14; ICEKp5</i>	140	KL2	<i>fepG, entS</i>	non synonymous mutation at 82 bp (T > G .. Cys 28 Gly)	+	+	-
24	ST14	-	0	KL2	<i>fepG, entS</i>	nil	+	+	-
25	ST14	-	0	KL2	<i>fepG, entS</i>	nil	+	+	-
26	ST14	<i>ybt 14; ICEKp5</i>	140	KL2	<i>fepG, entS</i>	nil	+	+	-
28	ST14	-	0	KL2	<i>fepG, entS</i>	nil	+	+	-
7	ST147	<i>ybt 9; ICEKp3</i>	183	KL64	<i>fepG, entS</i>	lack of <i>mgrB</i>	+	+	-
9	ST147	<i>ybt 9; ICEKp3</i>	183	KL64	<i>fepG, entS</i>	disrupted by IS110 element at 74th bp	+	+	-
16	ST147	-	0	KL10	<i>fepG, entS</i>	nil	+	+	-
17	ST147	-	0	KL10	<i>fepG, entS</i>	nil	+	+	-
27	ST147	-	0	KL10	<i>fepG, entS</i>	nil	+	-	-
19	ST231	<i>ybt 14; ICEKp5</i>	145	KL51	<i>fepG, entS</i>	disrupted by IS110 element at 116th bp	+	+	+
30	ST231	<i>ybt 14; ICEKp5</i>	145	KL51	<i>fepG, entS</i>	disrupted by IS110 at 117th bp	+	+	+
14	ST2096	<i>ybt 14; ICEKp5</i>	140	KL64	<i>fepG, entS, iucABCD</i>	nil	+	-	+
15	ST25	<i>ybt 14; ICEKp5</i>	149	KL2	<i>fepG, entS</i>	nil	+	+	-

Tree scale: 0.1

mgrB_mutations

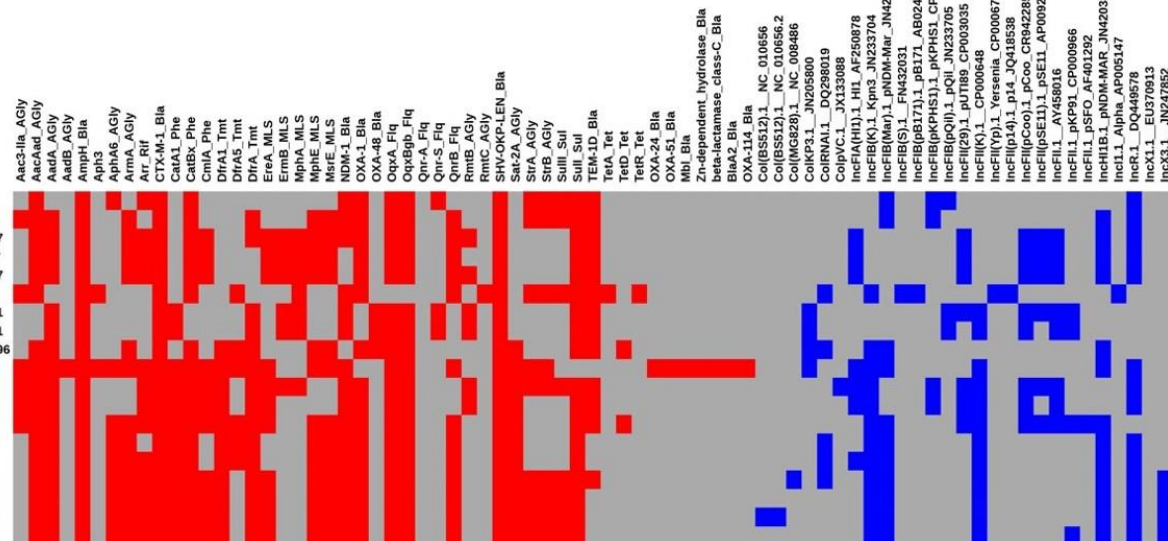
- no mutations
- deletion
- disruption by IS110
- disruption by ISKpn26
- Cys28Gly

Capsular Type

- KL2
- KL10
- KL51
- KL64

KL type

- 7 ST147
- 9 ST147
- 17 ST147
- 27 ST147
- 16 ST147
- 15 ST25
- 30 ST231
- 19 ST231
- 14 ST2096
- Ref
- 21 ST14
- 1 ST14
- 13 ST14
- 26 ST14
- 2 ST14
- 25 ST14
- 24 ST14
- 20 ST14
- 28 ST14



Reference	24361	24383	24288	24368	24273	22106	22122	1416	22934	84	107	70	68	71	69	72	107	1230	0
28 ST14	23506	23508	23555	23547	23539	21715	21732	2476	22438	1245	1248	1212	1169	1231	1172	1184	1248	0	1230
26 ST14	24435	24457	24362	24442	24347	22173	22189	1449	22997	121	2	85	83	108	84	87	0	1248	107
25 ST14	24400	24422	24327	24407	24312	22141	22157	1412	22970	86	87	50	16	73	17	0	87	1184	72
24 ST14	24378	24400	24305	24385	24290	22115	22131	1398	22951	83	84	47	3	70	0	17	84	1172	69
21 ST14	24390	24412	24317	24397	24302	22127	22143	1417	22951	51	108	70	69	0	70	73	108	1231	71
20 ST14	24397	24419	24324	24404	24309	22135	22151	1408	22959	82	83	46	0	69	3	16	83	1169	68
13 ST14	24398	24420	24325	24405	24310	22136	22152	1411	22960	84	85	0	46	70	47	50	85	1212	70
2 ST14	24431	24453	24358	24438	24343	22171	22187	1438	22991	121	0	85	83	108	84	87	2	1248	107
1 ST14	24393	24415	24320	24400	24304	22132	22148	1430	22957	0	121	84	82	51	83	86	121	1245	84
15 ST25	23303	23304	23189	23207	23172	23902	23917	23263	0	22957	22991	22960	22959	22951	22951	22970	22997	22438	22934
14 ST2096	23059	23061	24051	24118	24036	22215	22231	0	23263	1430	1438	1411	1408	1417	1398	1412	1449	2476	1416
30 ST231	24716	24742	24501	24579	24485	3	0	22231	23917	22148	22187	22152	22151	22143	22131	22157	22189	21732	22122
19 ST231	24684	24710	24469	24547	24453	0	3	22215	23902	22132	22171	22136	22135	22127	22115	22141	22173	21715	22106
27 ST147	1958	1979	6	105	0	24453	24485	24036	23172	24304	24343	24310	24309	24302	24290	24312	24347	23539	24273
17 ST147	1975	2048	101	0	105	24547	24579	24118	23207	24400	24438	24405	24404	24397	24385	24407	24442	23547	24368
16 ST147	1955	1975	0	101	6	24469	24501	24051	23189	24320	24358	24325	24324	24317	24305	24327	24362	23555	24288
9 ST147	1	0	1975	2048	1979	24710	24742	23061	23304	24415	24453	24420	24419	24412	24400	24422	24457	23508	24383
7 ST147	0	1	1955	1975	1958	24684	24716	23059	23303	24393	24431	24398	24397	24390	24378	24400	24435	23506	24361

SNP distance

